

Exposé zur Master's Qualification Paper

AI-driven Impact Measurement and Management: Developing and Evaluating a Framework using Inluma as a Case Study

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1 Introduction and Motivation

Measuring and managing social and environmental impact is a critical challenge for organizations operating in the public value and social innovation space. Existing approaches to Impact Measurement and Management (IMM) often suffer from fragmentation, lack of scalability, and limited data integration.

Advances in Artificial Intelligence (AI), particularly in Natural Language Processing (NLP) and Machine Learning (ML), open new opportunities for automating and improving the reliability of IMM. As CTO of the Public Value Hub, Leipzig and responsible for the development of the *Inluma* IMM tool, this work explores how AI can support IMM in practice.

2 Problem Statement and Research Gap

While frameworks such as SROI, IRIS+, or SDG-based metrics provide standards for evaluating impact, they rely heavily on Human In The Loop (HITL), qualitative data collection and interpretation. This creates challenges of comparability, transparency, and efficiency.

The research gap lies in the limited integration of AI methods in IMM processes. There is a need for validated frameworks that combine AI-driven analytics with established IMM practices to support decision-making in social enterprises.

3 Research Questions and Objectives

The central research question is:

How can Artificial Intelligence support and enhance Impact Measurement and Management in social enterprises through an applied framework?

Sub-questions include:

- Which AI methods are most suitable for evaluating qualitative and quantitative impact data?
- How can an AI-driven IMM framework improve efficiency, comparability, and transparency in impact assessment?
- What are the practical implications of applying such a framework in the case of Inluma?

4 Methodology

The research will be conducted in three phases:

1. **Literature Review:** Analysis of academic and practical work on IMM frameworks and AI applications.
2. **Case Study:** Application of IMM concepts to the development of Inluma.
3. **Prototype and Evaluation:** Design of a proof-of-concept AI-supported IMM tool and assessment through user feedback and benchmarking.

5 Expected Contribution

This work aims to:

- Contribute to academic literature by bridging applied AI research with IMM practices.
- Provide a validated framework for AI-driven IMM applicable to social enterprises.
- Deliver practical insights for the further development of *Inluma*.

6 Preliminary Outline

1. Introduction and Motivation
2. Theoretical Background: IMM frameworks and AI methods
3. Methodology
4. Case Study: Inluma
5. Results and Discussion
6. Conclusion and Outlook

References

- Ebrahim, A., & Rangan, V. K. (2014). What impact? A framework for measuring the scale and scope of social performance. *California Management Review*.
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- OECD (2019). *Impact Measurement Approaches*.